



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)
UNIT TEST
30-05-2021

JEE(Main + Advanced) : NURTURE COURSE (PHASE : I,I(B),II(A) & NURTURE (LIVE) I,I(B),II(A))

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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Topic :

PHYSICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

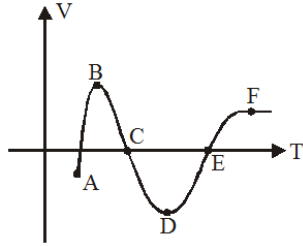
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

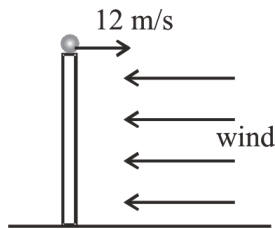
1. If $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = a\hat{i} + 3\hat{j}$ are perpendicular to each other. The value of a is :-
 (A) -4 (B) 4
 (C) -3 (D) 3
2. Maximum and minimum magnitude of resultant of two vectors of magnitude 'p' and 'q' are in the ratio 3 : 1; which of following relation is true ($p > q$) ?
 (A) $p = 2q$
 (B) $p = q$
 (C) $pq = 1$
 (D) None of these
3. If $\hat{A} = 0.5\hat{i} + 0.8\hat{j} + C\hat{k}$, then the value of C is:-
 (A) 0
 (B) 0.01
 (C) $\sqrt{0.11}$
 (D) Any value of C

4. A bird starts from (1, 0, 0) and moves in the direction $(2\hat{i} + 3\hat{j} - 6\hat{k})$ with a speed 21 m/s for 5 sec, then along the direction $(3\hat{i} + 4\hat{j} + 5\hat{k})$ with a speed $5\sqrt{2}$ m/s for next 5 sec. Find the final displacement of the bird so that it reaches the origin ?
 (A) $30\hat{i} + 45\hat{j} - 45\hat{k}$
 (B) $-45\hat{i} - 65\hat{j} + 65\hat{k}$
 (C) $-46\hat{i} - 65\hat{j} + 65\hat{k}$
 (D) $46\hat{i} - 65\hat{j} - 65\hat{k}$
5. Select incorrect statement for three vectors $\vec{a} = -3\hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} - 4\hat{k}$
 (A) Angle between vectors $\vec{a}$ and $\vec{b}$ is obtuse
 (B) Vectors $\vec{a}, \vec{b}, \vec{c}$ form a right angled triangle
 (C) Vectors $\vec{a}, \vec{b}, \vec{c}$ form an acute angled triangle
 (D) $\vec{a} \cdot \vec{c} = 0$ and $a^2 + c^2 = b^2$
6. A 10 gm bullet moving at 100 m/s strikes a wooden block kept fixed. The bullet comes to rest & stops in 10 cm. What is the average force (in N) experienced by the bullet ?
 (A) 250 N (B) 500 N
 (C) 100 N (D) 5000 N

7. A particle can only move in one dimension along the x-axis. The velocity of the particle as a function of time is shown in the diagram. There is a net force in the positive direction at

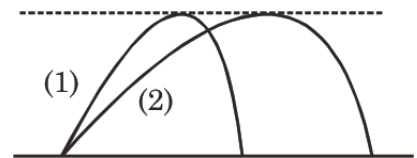


- (A) B (B) E (C) C (D) D
8. A body returns to the starting point after 20s. If the time rate of change of velocity during the entire motion is constant, its location after 13 s will be same as that after
- (A) 3s (B) 7s (C) 9s (D) 4s
9. Two cars start off to race with velocities 2m/s and 4m/s travel in straight line with uniform acceleration 2m/s^2 and 1m/s^2 respectively. The length of the path if they reach the final point at the same time is :-
- (A) 24 m (B) 12 m
(C) 5 m (D) Data insufficient
10. A ball is projected horizontally from the top of tower at 12 m/s as shown in figure. Wind is blowing as indicated (horizontally), which causes acceleration 6m/s^2 horizontally. Displacement of the ball when it strikes the tower, is ($g = 10\text{m/s}^2$)



- (A) 20m (B) 60 m (C) 80 m (D) 40 m

11. A body when projected vertically up, covers a total distance D during its time of flight. If there were no gravity, the distance covered by it during the same time is equal to :-
- (A) 0 (B) D
(C) 2D (D) 4D
12. A small ball is released from rest and it accelerates due to gravity, after 1 sec suddenly gravity disappears and again appears after 5 sec. Ball takes total 8 sec to reach the ground. The height above ground from where ball was released, is :-
- (A) 100 m (B) 95 m
(C) 320 m (D) 160 m
13. An astronaut in a strange planet observe that he can jump a maximum horizontal distance of 2m, if his initial speed is 6 m/s. What is the acceleration due to gravity of the planet ?
- (A) 3m/s^2 (B) 9m/s^2
(C) 18m/s^2 (D) 36m/s^2
14. Path of projectile of two particles is shown in the figure. If T_1 and T_2 are time of flight of particle (1) and (2) respectively, then :-



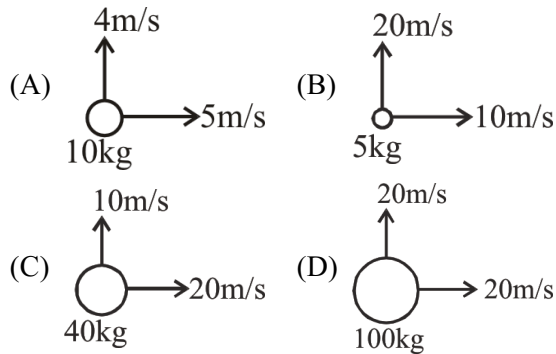
- (A) $T_1 > T_2$
(B) $T_1 < T_2$
(C) $T_1 = T_2$
(D) Can't say

15. A ball is thrown at an angle θ such that its range "R" is related to its time of flight "T" as $R = 5T^2$. Then the value of θ is :-

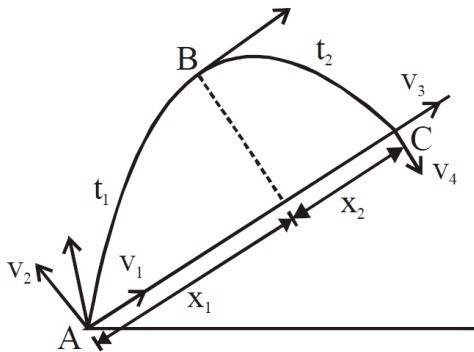


- (A) 45° (B) 90°
(C) 30° (D) 60°

16. Four balls are projected from ground as shown. Which will have maximum range :-

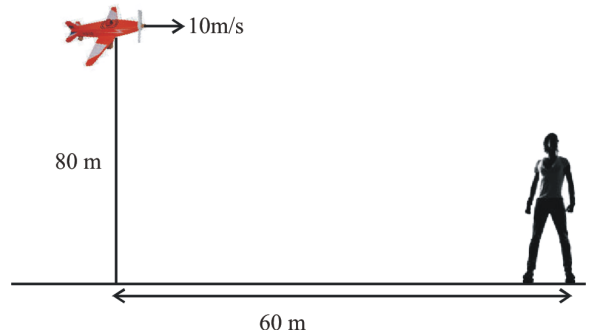


17. A ball is projected up on an inclined plane. If t_1 and t_2 are time taken from A to B and B to C then which of the following is wrong : (B is at maximum distance from inclined plane)



- (A) $t_1 = t_2$ (B) $x_1 = x_2$
(C) $v_1 > v_3$ (D) $|v_2| = |v_4|$

18. A toy plane is flying horizontally at 10 m/s at height 80 m above the ground towards a man on the ground as shown. When plane is 60m away horizontally from man, a ball is dropped by the plane, find speed of man (assumed uniform) so that he is able to catch the ball just before it strikes the ground :- ($g = 10 \text{ m/s}^2$)



- (A) 10 m/s (B) 15 m/s
(C) 5 m/s (D) 7.5 m/s

19. Two arrows are launched at the same time with the same speed. Arrow A at an angle greater than 45 degrees, and arrow B at an angle less than 45 degrees. Both land at the same spot on the ground. Which arrow arrives first?

- (A) arrow A arrives first
(B) arrow B arrives first
(C) they both arrive together
(D) it depends on the elevation where the arrows are launched

20. A balloon rises from rest on ground with constant acceleration $\frac{2g}{3}$. A stone is dropped when the balloon rises to a height 30m. The total time of flight after it is dropped is :

- (A) 2s (B) $3\sqrt{10}$ s (C) 3s (D) $2 + \sqrt{10}$ s

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

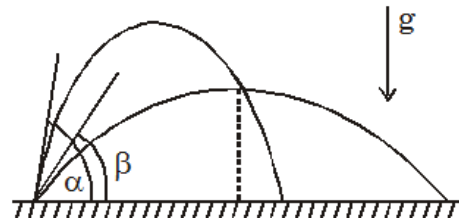
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

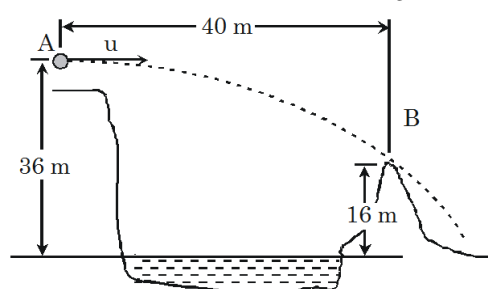
Zero Marks : 0 In all other cases.

1. A man starts from (4m, 3m, 2m) and travels along $(2\hat{i} + 2\hat{j} + \hat{k})$ with velocity 30 m/s for 5 sec. If final position of the man is (x, y, z) in meter, then find the value of (x + y + z).
2. The work done by a force $\vec{F}$ acting on a particle is given by $W = \vec{F} \cdot \vec{S}$ where $\vec{S}$ is displacement of the particle. If $\vec{F} = (\hat{i} + 2\hat{j})$ newton and $\vec{S} = (3\hat{i} + 2\hat{j})$ meter. Then find the Work done in joule.
3. If $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 7\hat{i} + 24\hat{j}$. A vector having same magnitude as $\vec{A}$ and parallel to $\vec{B}$ is given as : $\alpha\hat{i} + \beta\hat{j}$ the value of $\left(\frac{\alpha + \beta}{31}\right) 5$ is.
4. A ball of mass 100 g is moving with a velocity $\vec{v} = (2\hat{i} + 2\hat{j} + 2\hat{k})$ m/s. What is its kinetic energy in joule.
5. Velocity of particle depend on x as $v = 4x^2 - 4$ where x is position of the particle then find acceleration at x = 1 is.

6. From the hole of the hose, covered with a finger, two jets are coming out at angle $\alpha = 68^\circ$ and $\beta = 37^\circ$ to the horizontal with the same initial speed $v = 31.5$ m/s. The jet spray lies in the same plane. At what horizontal distance (in m) from the hose does the jet intersect? [$\sin 68^\circ = 12/13$]



7. The velocity of an object is given by the equation $v = At^3 - Bt$. Here 't' refers to time. Find the time at which the object is in equilibrium. Assume : $A = \frac{2}{3}$, $B = 50$.
[Hint : Force = Mass $\times$ Acceleration]
8. A body covers one-third of the distance with speed v_1 . For the remaining distance, it travels for half of the time with speed v_2 & another half of the time with speed v_3 . Find its average speed. If average speed is $\frac{nv_1(v_2 + v_3)}{4v_1 + v_2 + v_3}$. Fill n in OMR Sheet
9. A particle is projected at an angle 73° from horizontal with speed 22 m/s then find at what speed (in m/s) will it strike with ground ?
10. A boy throw a rock from A with minimum horizontal velocity 'u' so that it just clears the obstruction at B. Find the value of $\frac{u}{5}$.



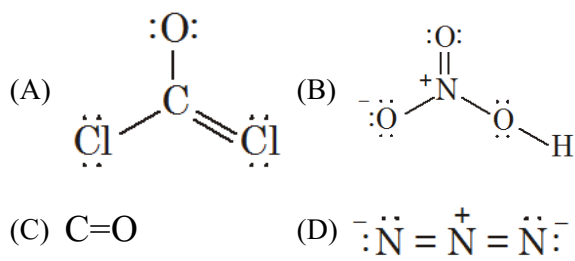
Topic :

CHEMISTRY

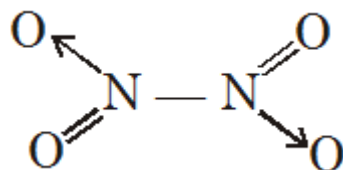
SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

1. Which of the following is/are **CORRECT** representation of Lewis structure :-



2. If following structure lies in XY plane, then select incorrect option/s.



- (A) All N-O bond lengths are equal
 (B) Nodal plane of all $N-\pi-O$ bonds lie in XY plane
 (C) Bond order of all N-O bonds is equal to 1.33
 (D) N-N bond has no nodal plane
3. The formal charge on central atom in Lewis dot structure of $(SO_4)^{2-}$ is equal to :-
 (A) +6 (B) -2 (C) +2 (D) 0

4. Which of the following overlap along x-internuclear axis results in δ -bond formation?



5. Atom X, Y, Z are oxygen, sulphur and chlorine respectively then choose correct statement(s) –

- (A) $X \longrightarrow Y \longrightarrow Z$ sequence is correct for increasing electron affinity
 (B) For electronegativity increasing sequence will be $Y \longrightarrow X \longrightarrow Z$
 (C) For $I.E_1$ increasing sequence will be $X \longrightarrow Z \longrightarrow Y$
 (D) $X \longrightarrow Y \longrightarrow Z$ sequence is increasing for both $I.E_1$ and EA_1 .

6. Match the list:

List- I (Order)		List- II (Properties)	
(P)	$Fe^{3+} > Fe^{2+}$	(1)	Radii
(Q)	$O > S$	(2)	Ionisation energy
(R)	$P > N$	(3)	Electron affinity
(S)	$I > I^+$	(4)	Electronegativity

The correct option is

- (A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$
 (B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$
 (C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$
 (D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

7. Select the **INCORRECT** statement(s) ?
- (A) The third ionisation energy of carbon is less than that of third ionisation energy of Beryllium.
- (B) The first ionisation energy of C^{2+} ion is greater than that of first ionisation energy of N^{2+} ion
- (C) The second ionisation energy of Mg is greater than the second ionisation energy of Na
- (D) The second ionisation energy of nitrogen is less than that of second ionisation energy of fluorine
8. Within each pair of elements of Sc & Zn, La & Lu and Al & Ga, respectively, the larger elements are -
- (A) Sc, La and Al (B) Zn, La and Ga
- (C) Sc, Lu and Ga (D) Zn, La and Al
9. The correct statements among I to III regarding hydration are,
- (I) The extent of hydration is more in aluminium ion than that of chloride ion
- (II) Hydration energy is most exothermic in the case of lithium ion as compare to all alkali metal ions
- (III) The hydrated radius is more in aquated sodium ion as compare to aquated magnesium ion
- (A) (I), (II) and (III) (B) (II) and (III)
- (C) (I) and (III) (D) (I) and (II)
10. The radius of Mn is minimum in which of the following compounds ?
- (A) Mn_2O_7 (B) MnO
- (C) MnF_4 (D) MnO_2

11. Which of the following statement is/are **INCORRECT**?
- (A) A cation is smaller than an anion if they are isoelectronic
- (B) Out of second period anion, fluoride ion is large in size
- (C) Both Mn^{2+} and Co^{2+} are paramagnetic
- (D) Generally metallic radius of an element ('X') is higher than its covalent radius
12. The specie which has $x = \text{product of } y \text{ and } z$, where $x = \text{number of ultimate shell electrons}$, $y = \text{number of penultimate shell electrons}$ and $z = \text{number of inner penultimate shell electrons}$ is :
- (A) Fe (B) Ni^{2+} (C) Ar (D) Cr^{2+}
13. The total number of electrons in Arsenic (As) for which the value of $(\ell + m) = 1$.
- (A) 15 (B) 9 (C) 7 (D) 11
14. Match the list:

	List-I (Atomic number)		List- II (If $\ell = (0 \text{ to } n)$ keeping all other rules same, the last electron will enter in _ _ _ _)
(P)	7	(1)	2d
(Q)	17	(2)	1p
(R)	21	(3)	3s
(S)	19	(4)	2p

The correct option is

- (A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 1$
- (B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$
- (C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$
- (D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

15. The general electronic configuration of outer most and penultimate shell is given as $(n-1)s^2 (n-1)p^6 (n-1)d^x ns^2$. Then for an element with $n = 4$ and $x = 7$.
The number of unpaired electron in the trivalent cation of the given element in isolated gaseous state is
(A) 0 (B) 3
(C) 4 (D) 1
16. **CORRECT** order of number of planes present in molecule which contain maximum number of atoms of corresponding molecule :
(A) $IF_7 > SF_6 > PCl_5$
(B) $PCl_5 > SF_6 > IF_7$
(C) $SF_6 > IF_7 > PCl_5$
(D) $PCl_5 > IF_7 > SF_6$
17. In the following reaction :
 $XeF_4 + F^- \rightarrow XeF_5^-$
Select the **incorrect** statement -
(A) More 'd' orbitals are involved in hybridisation in product compared to reactants
(B) % s character of hybrid orbital changes from reactant to product
(C) Both reactant and product idealised bond angle as per hybridisation
(D) "Maximum atoms lying in same plane" is same for XeF_4 & XeF_5^-
18. Which of the following species has / have same number of σ (sigma) & π (pie) bond ?
(A) $CN^\ominus$
(B) $XeOF_2$
(C) N_2
(D) $NO_2^\oplus$
19. Which of the following overlapping results in π -bond formation ?
(A) $p_x + d_{xy} \xrightarrow{z\text{-axis}}$
(B) $d_{xy} + d_{xz} \xrightarrow{y\text{-axis}}$
(C) $d_{xz} + d_{yz} \xrightarrow{y\text{-axis}}$
(D) $d_{yz} + p_z \xrightarrow{y\text{-axis}}$
20. Select correct about CO_3^{2-} carbonate ion in one of the lewis structure based on the presence of two single bonds and one double bond between carbon and oxygen atoms :
(A) Total number of lone pair = 8
(B) Formal charge on two oxygen = -1 and one oxygen = zero
(C) Oxidation number of C = +4 & Formal charge on C = zero
(D) All are correct

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- Find the number of molecules in which atleast one non-axial d- orbital is involved in hybridisation :- XeF_5^- , IF_7 , XeF_6 , XeOF_5^- , SO_3F^-
- In how many given species, d_{z^2} -orbital is used in the hybridisation of underlined atom?
 PO_4^{3-} , XeF_2 , SO_2 , IOF_3 , ClF_3 , SeF_4 , ICl_4^- , XeF_5^-
- Find out the total number of hypervalent species is.
 BF_3 , ClF_3 , SF_4 , ICl_2^- , XeF_5^+ , SnCl_2 , I_3^+ ,
- The number of species, which have nearly equal to average X–O bond order = 1.5 is (where X = central atom C/N/O).
 CO_3^{2-} , NO_2^- , NO_3^- , NO_2^+ , O_3 , CO_2 , CO
- From the given compounds of 'X' number of compounds are more acidic than sulphur dioxide and 'Y' number of compounds are less acidic than sulphur dioxide then find the value of (X/Y) ?
 SO_3 , Cl_2O_7 , Na_2O , SiO_2 , BeO , B_2O_3 , Al_2O_3

- Find the value of expression $\frac{(x+z)}{(y+w)}$ for following oxides.
 B_2O_3 , OCl_2 , CO , PbO , Cr_2O_3 , MnO , HgO , Mn_2O_7 , Cs_2O
where
 x = total number of acidic oxides
 y = total number of basic oxides
 z = total number of amphoteric oxides
 w = total number of neutral oxides
- Assuming z axis is internuclear axis, find total number of combination of atomic orbitals, in which nodal plane of formed bond passes through internuclear axis.
 $p_x + p_x, p_z + p_z, d_{xy} + d_{xy}, d_{yz} + d_{yz}, d_{x^2-y^2} + d_{x^2-y^2}, p_y + d_{yz}, p_x + d_{xz}, d_{z^2} + d_{z^2}, p_z + d_{z^2}$
- Among the following, find out the total number of process associated with the formation of NaCl from elemental form of reactants.
Ionisation energy of sodium, electron affinity of sodium, electronegativity of chlorine, bond dissociation energy of chlorine, sublimation enthalpy of sodium, lattice energy of NaCl.
- Find the number of orbitals which can not form p-bond as well as d-bond both
 $s, p_x, p_y, p_z, d_{x^2-y^2}, d_{z^2}, d_{xy}, d_{yz}, d_{zx}$
- Which of the following specie(s) have anionic part PLANAR
 N_2 , $\text{O}_5(\text{s})$, $\text{PCl}_5(\text{s})$, $\text{I}_2(\text{l})$, Cl_2 , $\text{O}_6(\text{s})$, NaBF_4 , $\text{HNO}_3(\text{aq.})$

Topic :

MATHEMATICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

1. If α and β are solutions of $x^2 - 10x + 10 = 0$ such that $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = b$, then roots of equation $2x^2 + 10x + 3b = 0$ are
 (A) rational and distinct
 (B) real and equal
 (C) imaginary
 (D) irrational and distinct
2. If equations $x^2 - bx + a = 0$, $x^2 + ax - b = 0$ have exactly one common root, then sum of their uncommon roots is
 (A) 1 (B) 0 (C) -1 (D) $a + b$
3. If both roots of the equation $x^2 - 6ax + 2 - 2a + 9a^2 = 0$ exceed 3, then
 (A) $a < 1$ (B) $a > \frac{11}{9}$
 (C) $a > \frac{3}{2}$ (D) $a < \frac{5}{2}$
4. If α, β, γ are roots of the cubic equation $x^3 + 2x - 1 = 0$ then value of $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$ is
 (A) 1 (B) -1 (C) 2 (D) -2

5. The smallest value of $x^2 - 3x + 3 \forall x \in [-3, 5/2]$, is
 (A) $3/4$ (B) 5 (C) -15 (D) -20
6. An equation $(2x^2 + 1 - 3x)(2x^2 + 1 + 5x) - 9x^2 = 0$ has
 (A) two real and two imaginary roots
 (B) all imaginary roots
 (C) four real roots
 (D) all roots are equal
7. All values of m for which $mx^2 - 6mx + 5m + 1 > 0$ for all real x , is
 (A) $0 \leq m < \frac{1}{4}$ (B) $m \geq \frac{1}{4}$
 (C) $m < 0$ (D) None of these
8. The expression $2x^2 + 4xy - ky^2 + 4x + 2y - 1$ can be resolved into two linear factors, then the value of k is -
 (A) $-\frac{5}{3}$ (B) $\frac{5}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$
9. Roots of the equation $(x - 2)(x - 4) + \lambda(x - 1)(x - 3) = 0$, where $\lambda > 0$, are
 (A) Real and distinct
 (B) Real and equal
 (C) Imaginary
 (D) One real and one imaginary

10. If α & β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then the roots of quadratic equation $ax^2 - bx(x-1) + c(x-1)^2 = 0$ are
- (A) $\frac{\alpha}{1-\alpha}, \frac{\beta}{1-\beta}$
 (B) $\alpha - 1, \beta - 1$
 (C) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$
 (D) $\frac{1-\alpha}{\alpha}, \frac{1-\beta}{\beta}$
11. 1 lies between the roots of equation $-x^2 + ax + a = 0$ if and only if a lies in interval :-
- (A) $\left(\frac{1}{2}, \infty\right)$ (B) $\left[-\frac{1}{2}, \infty\right)$
 (C) $\left(-\infty, \frac{1}{2}\right)$ (D) $\left(-\frac{1}{2}, \frac{1}{2}\right)$
12. Sum of integral value(s) of x satisfying the inequality $\frac{x^2 - 3x - 10}{x^2 + x + 1} \leq 0$, is
- (A) 9 (B) 10
 (C) 12 (D) 15
13. Let α and β be the roots of the equation, $2x^2 + 3x - 1 = 0$. If $S_n = \alpha^n + \beta^n$, $n = 1, 2, 3, \dots$, then
- (A) $2S_7 + 3S_6 = S_5$ (B) $2S_7 + 3S_6 = 1$
 (C) $4S_7 + 3S_6 = 2S_5$ (D) None of these
14. The set of values of 'a' for which $ax^2 + (a-2)x - 2$ is negative for exactly two integral x , is
- (A) (0, 2) (B) [1, 2)
 (C) (-1, 2) (D) (0, 2]
15. If $\left| \frac{x^2 + mx + 1}{x^2 + x + 1} \right| < 3$ for all real x , then
- (A) $m < -1$ (B) $-1 < m < 5$
 (C) $m > 7$ (D) $m > 9$
16. Solution set of $\sqrt{-x^2 + x + 6} > (x-4)$ is -
- (A) $(-\infty, -2] \cup \{3\}$
 (B) $(-\infty, -2] \cup [3, 4)$
 (C) $(2, 5/2)$
 (D) $[-2, 3]$
17. If $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$ has no real roots and $a - b + c < 0$, then -
- (A) $a > 0$ (B) $c < 0$
 (C) $a = 0$ (D) $ac < 0$
18. If α, β are the roots of $ax^2 + bx + c = 0$, if $\alpha < 0 < \beta$ and $|\alpha| > |\beta|$, then
- (A) $a > 0, b > 0, c > 0$
 (B) $a > 0, b < 0, c > 0$
 (C) $a < 0, b < 0, c < 0$
 (D) $a < 0, b < 0, c > 0$
19. Maximum value of $\frac{x^2 + 4x + 7}{x^2 + 4x + 5}$ is
- (A) 3 (B) 4 (C) $\frac{17}{6}$ (D) $\frac{17}{4}$
20. Number of integers in range of $x^2 - 5|x| + 1$, $x \in [-4, 2]$ is
- (A) 7 (B) 5
 (C) 6 (D) 8

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

7. Find product of all real values of x satisfying $(5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10$
8. If '4' lies between the roots of the equation $x^2 - (3k - 1)x + 5k = 0$, then minimum possible integral value of k is
9. Sum of all real solution of the equation $|x - 3|^2 - |x - 3| - 2 = 0$ is
10. If a, b, c are positive real no.'s then the number of real roots of the equation $ax^2 + b|x| + c = 0$ is

1. If $(a^3 - 3a - 2)x^2 + (a^2 - a - 2)x = 2a^4 - 10a + k$, is true for all real values of x , then the value of k is
2. If all real values of x obtained from the equation $4^x - (a - 3)2^x + a - 4 = 0$ are non-positive, then $a \in (p, q]$, then $\frac{q}{p}$ is equal to
3. Number of integral values of 'a' such that $x^2 + ax + a + 1 = 0$ has integral roots is equal to
4. Sum of non real roots of equation $(x^2 + x - 2)(x^2 + x - 3) = 12$ is
5. Number of integers in range of $\frac{2x+3}{x+4}$, $x > 1$ is
6. Number of integral values of 'b' for which inequality $(a^2 + 1)x^2 + 4(a + b)x + 2 < 0$ is true for atleast one 'x' & $\forall a \in \mathbb{R}$, is -

Space for Rough Work